SSB $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{EM}}$ from T^4/Z_3 Geometry

Spontaneous Symmetry Breaking as Topological Consequence

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7 September 2026

Contents

Status Markers

[K] *Confirmed* — numerically verified.

[B] *Proven* — algebraically demonstrated.

[S] *Speculative* — open.

[X] *Experimental* — confirmed.

[Q] *Source* — external primary value.

Abstract

The spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$ is implemented in the Standard Model through the Higgs potential $V(\Phi)$ with $\mu^2 < 0$ — a postulate. This document derives it from three building blocks of the T^4/Z_3 geometry, all already proven in the corpus.

The three building blocks:

1. $\tilde{T} \cdot m = 1$ forces $\langle\Phi\rangle \neq 0$: the vacuum must carry mass **[B]** (Docs. 003, 010, 306, 334).
2. The Z_3 Galois structure forces $Q_{\text{vac}} = 0$: the vacuum is electrically neutral **[B]** (Doc. 346).
3. $Q = 0$ of the vacuum selects $U(1)_{\text{EM}}$ as the residual group: the Gell-Mann-Nishijima generator annihilates the vacuum **[B]** (Doc. 347).

Main finding: The symmetry-breaking structure is topologically forced, not postulated **[B]**. What appears in SM convention as the spontaneously breaking postulate $\mu^2 < 0$ is in FFGFT a consequence of the time-field geometry and the Galois classification.

Open point [S]: CKM/PMNS mixing angle values (Doc. 348 Theorem D). New candidate: GF(3^6) = GF(729) and 7th roots of unity (Matzke IPI mail, 7 September 2026, R115).

Overview: All SM fermion quantum numbers and their certificate status are summarised in Doc. 356 (particle diagram) and Doc. 357 (certificate table).

1 Background

The SM Higgs Potential

The Standard Model implements SSB through:

$$V(\Phi) = \mu^2 |\Phi|^2 + \lambda |\Phi|^4, \quad \mu^2 < 0, \tag{1}$$

with minimum $|\Phi|^2 = -\mu^2/(2\lambda) = v^2/2$, $v \approx 246\,\mathrm{GeV}$. The Higgs doublet $\Phi = (\phi^+, \phi^0)^T$ has $I_3 = (-\frac{1}{2}, +\frac{1}{2})$, $Y = +1$. The neutral component develops the VEV:

$$\langle\Phi\rangle = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix}, \quad Q_{\text{vac}} = I_3 + \frac{Y}{2} = -\frac{1}{2} + \frac{1}{2} = 0. \tag{2}$$

The surviving symmetry is $U(1)_Q$ with generator $Q = I_3 + Y/2$ — this is $U(1)_{\text{EM}}$ **[X]** (PDG **[Q]**).

The question: Why does the vacuum have $Q = 0$? Why $\mu^2 < 0$? In the SM: postulated. In FFGFT: from the geometry.

2 Building Block 1: VEV $\neq 0$ from $\tilde{T} \cdot m = 1$

The time-field duality (Docs. 003, 010) in natural units:

$$\tilde{T}(x,t) \cdot m(x,t) = 1. \tag{3}$$

This relation has no solution at $m = 0$: $\tilde{T} \to \infty$ is not a physical state (Docs. 306, 334). The vacuum must carry a non-vanishing mass:

$$m_{\text{vac}} \neq 0 \quad \Longleftrightarrow \quad \langle\Phi\rangle \neq 0. \tag{4}$$

Connection to the Higgs potential: In SM language, $m_{\text{vac}} = 0$ is the minimum of V for $\mu^2 > 0$ — an empty vacuum. For $\mu^2 < 0$ the minimum lies at $|\Phi| \neq 0$. The condition (4) from FFGFT is equivalent to the requirement $\mu^2 < 0$ in the SM potential: both force $\langle\Phi\rangle \neq 0$ **[B]**.

The difference: In the SM, $\mu^2 < 0$ is a postulate. In FFGFT, $m_{\text{vac}} \neq 0$ follows from the reciprocity $\tilde{T} \cdot m = 1$, which itself follows from $E = mc^2$ (Doc. 354 Insight 1) **[B]**.

3 Building Block 2: $Q_{\text{vac}} = 0$ from the Galois Structure

Step 2a: The vacuum is a torus background

The time field in the vacuum is $\tilde{T}_0 = 1/m_0 = \text{const}$ — a static, spatially homogeneous configuration. In the T^4/Z_3 framework this is the $k = 0$ mode (zero winding): no angular momentum, no colour charge, no electric charge. The vacuum is topologically trivial **[B]** (Doc. A075).

Step 2b: Galois classification of the vacuum

In GF$(27)^*$, the combination of QR-property mod 13 and N mod 3 colour charge classifies all fermions (Doc. 346 Theorem B **[B]**):

QR mod 13	N mod 3	Particle type	Q
QR (Quadratic Residue)	$\equiv 0$	Neutrino	0
NQR (Non-QR)	$\equiv 0$	Charged lepton	-1
NQR	$\equiv 1, 2$	Quark	$\pm 1/3, \pm 2/3$

The vacuum is no fermion — it is the $k = 0$ mode, the topologically trivial element. Its decisive property:

- No colour charge: $N \bmod 3 = 0$ (colour singlet) **[B]**
- Electrically neutral: QR-property, hence $Q = 0$ **[B]**

This follows from Doc. 346 Theorem A **[B]**: $Q = 0 \iff$ QR mod 13. The topologically trivial element (zero winding) is QR **[B]**.

Step 2c: The vacuum has $I_3 = -1/2$, $Y = +1$ with $Q = 0$

The doublet structure follows from the Sd sector (Doc. 349 Theorem D **[B]**): the Sd sector (left-handed, odd Cl(6) grades) carries $I_3 = -1/2$. The hypercharge Y follows from the Legendre symbol (Doc. 347 Theorem A **[B]**). The vacuum in the Sd sector with $I_3 = -1/2$, $Y = +1$:

$$Q_{\text{vac}} = I_3 + \frac{Y}{2} = -\frac{1}{2} + \frac{1}{2} = 0. \tag{5}$$

$Q_{\text{vac}} = 0$ is algebraically forced **[B]**.

4 Building Block 3: $U(1)_{\text{EM}}$ as Residual Group

Residual symmetry of the vacuum

The symmetry group $G = SU(2)_L \times U(1)_Y$ acts on the vacuum. The residual group $H \subset G$ after breaking is the subgroup that leaves the vacuum invariant:

$$H = \{g \in G \mid g|\text{vac}\rangle = |\text{vac}\rangle\}. \tag{6}$$

The generator $Q = I_3 + Y/2$ (Gell-Mann-Nishijima, Doc. 347 Theorem C **[B]**) annihilates the vacuum:

$$Q|\text{vac}\rangle = Q_{\text{vac}}|\text{vac}\rangle = 0. \tag{7}$$

The one-parameter subgroup:

$$U(1)_Q : \quad e^{i\alpha Q}|\text{vac}\rangle = |\text{vac}\rangle \quad \forall\, \alpha \in \mathbb{R}. \tag{8}$$

$U(1)_Q$ leaves the vacuum invariant — this is $U(1)_{\text{EM}}$ **[B]**.

All other generators break the symmetry

The remaining generators of $SU(2)_L \times U(1)_Y$ generate the massive bosons $W^\pm$, Z^0 (Goldstone theorem **[X]**):

$$W^\pm \leftrightarrow T_\pm = T_1 \pm iT_2, \quad T_\pm|\text{vac}\rangle \neq 0 \tag{9}$$

$$Z^0 \leftrightarrow T_3 - \frac{Y}{2}\sin^2\theta_W, \quad Z^0|\text{vac}\rangle \neq 0. \tag{10}$$

The photon γ corresponds to the generator $Q = T_3 + Y/2$, which annihilates the vacuum — it remains massless **[B]**. The mass ratio $M_W/M_Z = \cos\theta_W$ follows from the Weinberg angle derived from ξ (Doc. 323 **[K]**).

5 Complete Chain

Step	Statement	Status	Source
1	$\tilde{T} \cdot m = 1$ from $E = mc^2$	**[B]**	Docs. 003,010,052
2	$m_{\text{vac}} \neq 0$: no massless vacuum	**[B]**	Docs. 306,334
3	Vacuum is $k = 0$ mode (topologically trivial)	**[B]**	Doc. A075
4	$Q_{\text{vac}} = 0$: trivial element is QR	**[B]**	Doc. 346 Thm. A
5	Sd sector: $I_3 = -1/2$ (left-handed)	**[B]**	Doc. 349 Thm. D
6	$Y = +1$ from Legendre symbol	**[B]**	Doc. 347 Thm. A
7	$Q = I_3 + Y/2 = 0$ (Gell-Mann-Nishijima)	**[B]**	Doc. 347 Thm. C
8	$U(1)_Q$ leaves vacuum invariant: residual group	**[B]**	this doc.
9	$W^{\pm}$, Z^0 massive; γ massless	**[B]**+**[X]**	Goldstone + Doc. 323
SSB $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$		**[B]**	complete

6 Numerical Consistency

Quantity	FFGFT	Experiment **[Q]**	Dev.	Status
$\sin^2 \theta_W(M_Z)$	0.2308	0.2312	$-0.17\,\%$	**[K]**
M_W/M_Z	$\cos \theta_W = 0.8770$	0.8814	$-0.50\,\%$	**[K]**
Q_{vac}	0 (exact)	0 (exact)	0	**[B]**
m_γ	0 (exact)	$< 10^{-18}$ eV	—	**[B]**+**[X]**

7 Open Points

CKM/PMNS mixing angle values [S]: The structure of the mixing matrices (which entries vanish, which are $1/\sqrt{2}$) is proven **[B]** (Doc. 348 Theorems B,C). The numerical values of the mixing angles require further algebraic work (Doc. 348 Theorem D). This is the only remaining **[S]** in the gauge/Higgs complex.

New candidate: GF(3^6) = GF(729) and 7th roots of unity [S]:

IPI correspondence with Douglas Matzke (7 September 2026) yields a new algebraic finding: 7th roots of unity require an extension to GF(3^6) = GF(729) **[B]**. The algebraic reason is compelling: $x^7 - 1$ factors over GF(3) as $(x-1) \cdot \phi_7$, where ϕ_7 is *irreducible of degree 6* — a degree-6 extension is therefore minimal and necessary.

GF(729)* has order $728 = 8 \times 7 \times 13$. The prime factors yield three scales:

- 13: GF(27) level — quantum numbers, charges, generations **[B]**
- 7: order-7 elements exist in G(6) over GF(9) (Doc. 341 §6, `pruef_341_root7_gf9`) **[B]**; their reading as *phases* $k \cdot 360°/7$ is an $\mathbb{R}$-interpretation — characteristic 3 has no angles, and cos($2\pi/7$) $\notin$ GF(3^k) for any k — hence **[S]**
- 8: octagonal structure; $360°/8 = 45°$ — near $\theta_{23}^{\text{PMNS}} \approx 49.2°$ (candidate **[S]**)

Concrete observation [S]: $\theta_{23}^{\text{PMNS}} \approx 49.2°$ lies $2.2°$ from $360°/7 \approx 51.4°$ — within measurement uncertainty, but not a proof. The CP phase $\delta_{CP}^{\text{CKM}} \approx 68°$ has no direct relation to 7th roots of unity; $\delta_{CP}^{\text{PMNS}} \approx 195° \approx 4 \times 360°/7 - 10°$ is likewise not a direct match.

What this means: GF(729) is the natural next step beyond GF(27). If CKM/PMNS angles reside at this level, Doc. 348 Theorem D would need to be executed on GF(729) — a concrete open task, not a structural obstacle.

Why P_+ and not P_- [S]: The chiral projection P_+ selects the left-handed Sd sector (A095). Why nature selects P_+ (and not P_-) is geometrically plausible but not derived from a deeper symmetry breaking **[S]**.

8 Verification Script

- Numerical verification of the SSB chain: $Q_{\text{vac}} = 0$, residual symmetry $U(1)_Q$, $W^{\pm}/Z^0$ mass ratio from θ_W, Goldstone counting, GF(729) order: `python/Dok355_Skripte/pruef_355_ssb.py`

9 Use of AI Tools

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